

Board exercises: score evolution across internal time u and diffusion time t

General goal. Investigate how the score function of a diffusion model evolves when the underlying (forward) process is fully known. We distinguish:

- an *internal time* (or trajectory index) u , describing the physical or semantic evolution of a dynamic object;
- a *diffusion time* t , describing independent Ornstein–Uhlenbeck (OU) corruption of each frame.

Dynamic object / trajectory. A trajectory is a sequence

$$a^\mu = \{a^\mu(u)\}_{u=0}^{T-1}, \quad a^\mu(u) \in \mathbb{R}^d, \quad \mu = 1, \dots, |\mathcal{D}|.$$

The distribution of clean trajectories induces, for each u , a marginal $p_0(\cdot | u)$.

Per-frame forward diffusion (Ornstein–Uhlenbeck (OU)). Conditioned on a clean frame $a(u)$, the corrupted observation at diffusion time t is

$$X_t(u) = a(u)e^{-t} + \sqrt{1 - e^{-2t}} Z(u), \quad Z(u) \sim \mathcal{N}(0, I_d),$$

independently for each u . Thus

$$p_t(x | a(u)) = \mathcal{N}(x; a(u)e^{-t}, (1 - e^{-2t})I_d), \quad p_t(x | u) = \int da p_0(a | u) p_t(x | a).$$

Per-frame score.

$$s(x, u, t) \equiv \nabla_x \log p_t(x | u).$$

1 Exercise 1 (left): 2D motion on a circle — discrete rotation and continuous limit

Discrete internal-time dynamics (as on the board)

Let (r_0, θ_0) be random with

$$r_0 \sim p_0(r), \quad \theta_0 \sim \text{Unif}[0, 2\pi).$$

where p_0 is Gaussian with mean r and variance 1. Define

$$a(0) = \begin{pmatrix} x_0 = r_0 \cos \theta_0 \\ y_0 = r_0 \sin \theta_0 \end{pmatrix}.$$

The internal-time evolution is given by a discrete rotation plus noise:

$$a(u+1) = R_{\Delta\theta} a(u) + \xi_u, \quad \xi_u \sim \mathcal{N}(0, \sigma_u^2 I_2),$$

where

$$R_{\Delta\theta} = \begin{pmatrix} \cos \Delta\theta & -\sin \Delta\theta \\ \sin \Delta\theta & \cos \Delta\theta \end{pmatrix}.$$

This generates a Markov chain on $\mathbb{R}^2$ whose marginals define $p_0(\cdot | u)$.

Continuous-time limit (intuition)

For small Δu , this discrete dynamics can be seen as a discretization of radial relaxation plus angular drift:

$$\begin{aligned} dr(u) &= (1 - r(u)) du + \sqrt{2T_r} dB_r(u), \\ d\theta(u) &= \omega du + \sqrt{2T_\theta} dB_\theta(u), \end{aligned}$$

but in this exercise the *discrete rotation picture is primary*.

Per-frame diffusion and exact score

For fixed u , the corrupted frame distribution at diffusion time step t is the product between the pdfs of the radius and the angle, i.e.

$$p_t(x | u) = \int_{\mathbb{R}^2} da p_0(a | u) \mathcal{N}(x; ae^{-t}, (1 - e^{-2t})I_2).$$

2 Exercise 2 (center): 1D generic one-step kernel

Internal-time transition

We consider a generic one-step Markov kernel

$$x_{u+1} = x_u + c + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2),$$

with no further structural assumptions. The joint density is

$$p(x_u, x_{u+1}) = p_0(x_u) \mathcal{N}(x_{u+1}; x_u + c, \sigma^2).$$

Joint score components

Define

$$\begin{aligned} S_0(x_u, x_{u+1}) &= \partial_{x_u} \log p(x_u, x_{u+1}) = \partial_{x_u} \log p_0(x_u) + \frac{x_{u+1} - (x_u + c)}{\sigma^2}, \\ S_1(x_u, x_{u+1}) &= \partial_{x_{u+1}} \log p(x_u, x_{u+1}) = -\frac{x_{u+1} - (x_u + c)}{\sigma^2}. \end{aligned}$$

These define a vector field on the (x_u, x_{u+1}) plane, as drawn on the board.

Marginal and per-frame score

The marginal is

$$p_0(x_{u+1}) = \int dx_u p_0(x_u) \mathcal{N}(x_{u+1}; x_u + c, \sigma^2),$$

and after OU corruption,

$$p_t(x | u) = \int da p_0(a | u) \mathcal{N}(x; ae^{-t}, 1 - e^{-2t}), \quad s(x, u, t) = \partial_x \log p_t(x | u).$$

3 Exercise 3 (right): fixed internal index, varying diffusion time

Single-slice view

Fix u and consider $\{p_t(\cdot | u)\}_{t \geq 0}$. The board sketches show how the density and score change as t increases.

Radial/elliptic example

Suppose

$$p_t(x | u) = f_t\left(\frac{\|x - m\|^2}{D_t}\right).$$

Then

$$s(x, u, t) = \nabla_x \log p_t(x | u) = \frac{f'_t\left(\frac{\|x - m\|^2}{D_t}\right)}{f_t\left(\frac{\|x - m\|^2}{D_t}\right)} \frac{2(x - m)}{D_t}.$$

This highlights how the score magnitude scales with the diffusion scale D_t .

Exact OU form

In general,

$$p_t(x | u) = \int da p_0(a | u) \mathcal{N}(x; ae^{-t}, (1 - e^{-2t})I_d), \quad s(x, u, t) = \nabla_x \log p_t(x | u).$$